

SRL-2018 Forensic Analysis Report

SRL-2018 — Digital Forensic & Incident Response Report

Case ID	SRL-2018-COMPROMISED-ENTERPRISE
Examiner	victor.galvan
Evidence	/cases/SRL-2018 — 7 disk (E01) + 22 memory (raw) images, 198 GB
Estate	Stark Research Labs (shieldbase.lan) — DC, file, mail, RD farm, workstations, DMZ-FTP
Incident window	2018-08-16 → 2018-09-05 UTC
Classification	Targeted intrusion / APT — intellectual-property theft
Status	10 findings APPROVED (examiner-signed, HMAC chain)

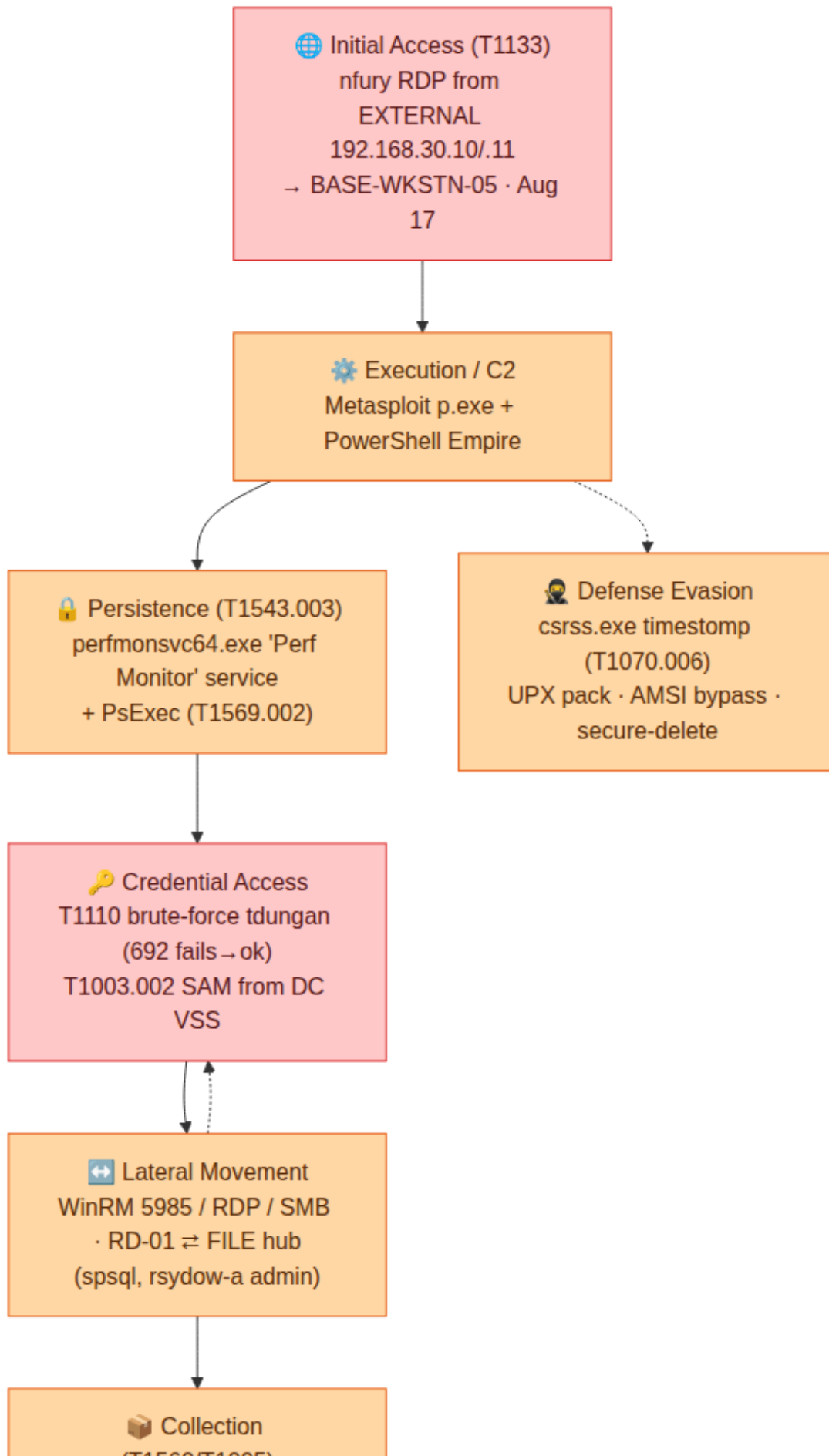
Evaluation note. The IP 42.112.153.164 was **operator-injected to test the system** and is **NOT** a real indicator — it has **zero evidentiary presence**. Disregard it in any tasking.

1. Executive summary

A targeted intrusion against the Stark Research Labs network, conducted with a **dual offensive framework — Metasploit/Meterpreter + PowerShell Empire** — culminating in the **theft of the "Carbonadium" research project** belonging to user nfury. The actor gained an external **RDP** foothold on a workstation, deployed a DNS-tunnelling implant (p.exe) and Empire agents across four hosts, **stole credentials** (NTLM brute-force + SAM-from-VSS), moved laterally via **WinRM/RDP/SMB** through an **RD-01 ≠ FILE** hub, **collected** the target IP with Rar/7-Zip, and **exfiltrated via the DMZ-FTP** server — covering tracks with a custom **timestamping** tool and thorough secure deletion.

The malware is fully recovered and hashed; the credential theft, lateral path, and objective are all evidenced.

2. Attack lifecycle (MITRE ATT&CK)



(1100/1100)
Rar.exe + 7za.exe
nfury 'Carbonadium' project



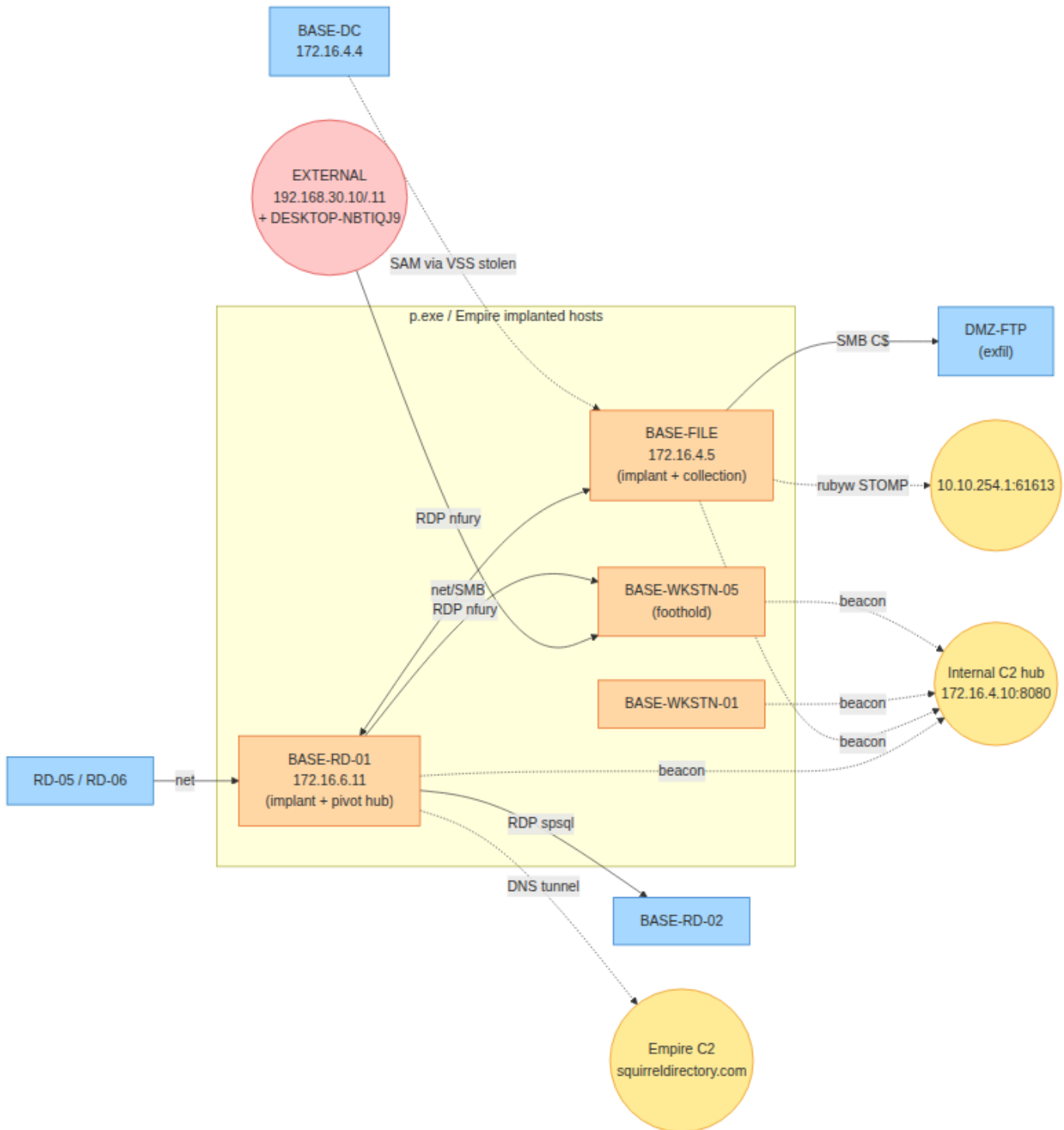
↑ Exfiltration
DMZ-FTP (C\$ staging)



🎯 OBJECTIVE
IP theft:
Carbonadium

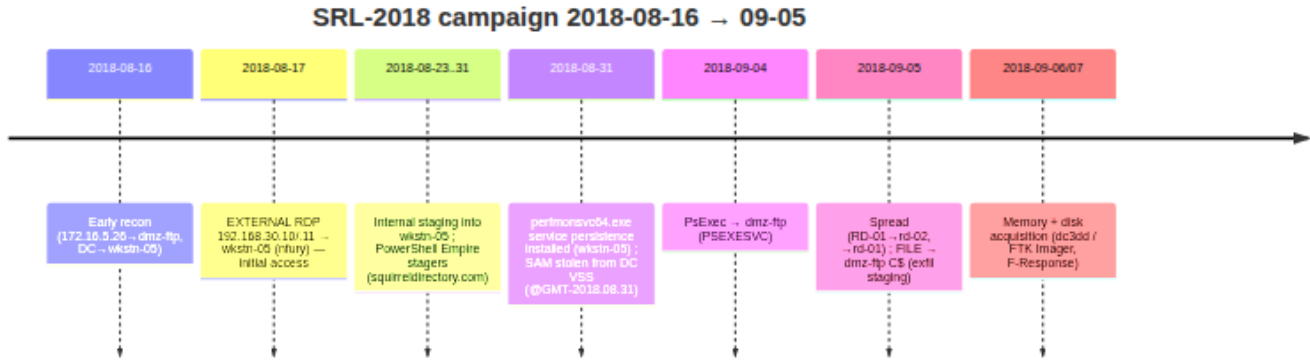


3. Lateral-movement & C2 architecture



Defenders excluded: BASE-HUNT / HUNT-02 / HUNT-03 (172.16.5.25/.27/.28) and BASE-ADMIN (172.16.5.26) are the **IR / threat-hunting** hosts (analyst cbarton = the acquisition examiner Clint Barton). Their fan-out logons are legitimate IR sweeps and are **not** attacker activity.

4. Timeline (UTC)



5. Recovered malware toolkit

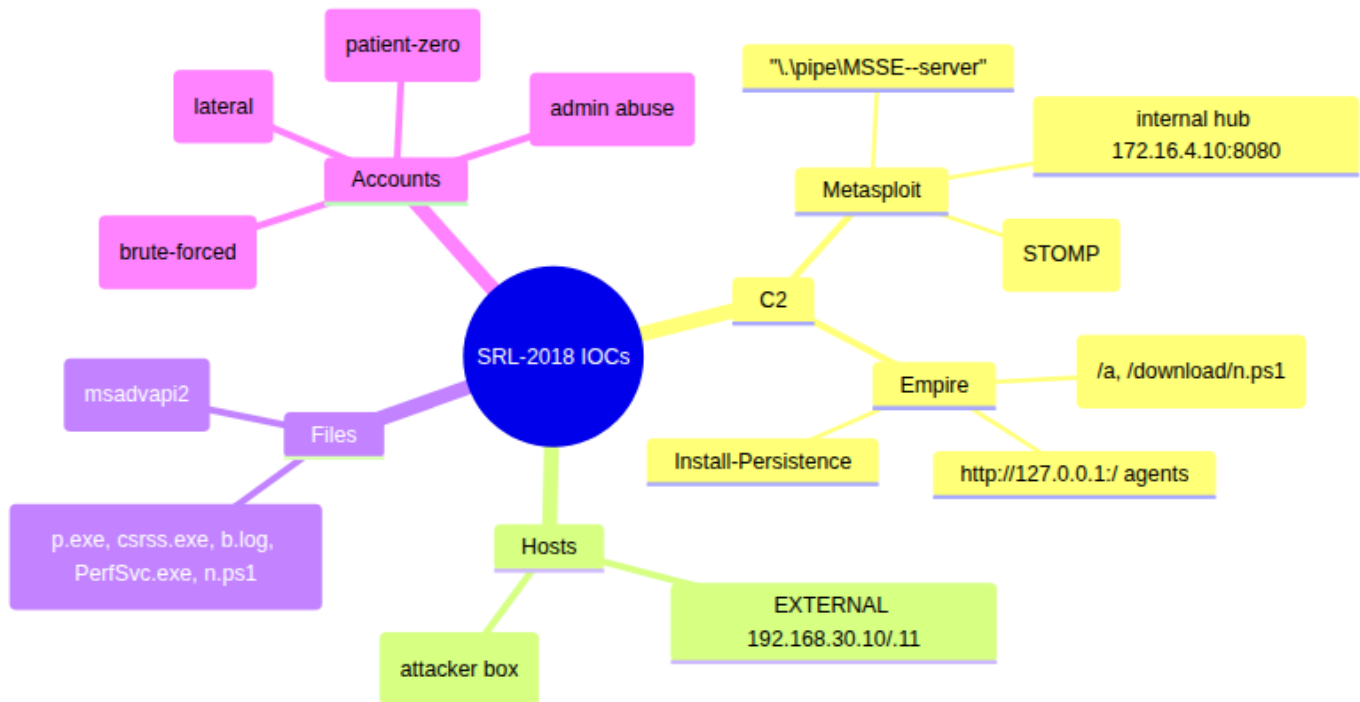
All samples recovered, hashed, and quarantined (defanged) at /home/admin2/srl-2018-malware-samples/.

File	SHA-256	Role	ATT&CK
p.exe (packed)	7fa4f6cc4e1bb27da7d9af7a2a533e72751b025b063e1df4359ebe127fd2892c	Metasploit/Meterpreter implant — DNS tunnel + \\.\pipe\MSSE-<N>-server; UPX	T1071.004, T1055, T1572
p.exe (unpacked)	d391ede758b6c769f89addb35ee9ec74eb0ae3a23831dbd6f7d932851265eee7	UPX-decompressed (reveals MSSE pipe)	—
perfmomsv64.exe	42477dd9317c739043d4516e04221743e00b737d1234f914a0e7608202758972	service persistence loader ("Perf Monitor")	T1543.003
csrss.exe	027fef173a71142cf1616e32290b9c52cbb425bfdeac1babd2a57e260c27c70e	timestomping tool (masquerades as csrss)	T1070.006, T1036
b.log	0a040d6f452063f5bef500972f83429f3654ada88eff13c2883e8eb06b519e05	BrowserHistoryView dump of nfury → objective	T1217, T1005
install_msadvapi2_32.exe	ebb75bbae3e1298cecbcd3c5b1b0ca0a2a8d4d17836f672546218bc47da8dc03	backdoor installer ("install_wormhole")	T1105
install_msadvapi2_64.exe	27d4968716a095a15956f1fb9e247b32dd7765b2e67149e17469908750280568	same (64-bit) — identical hash on FILE + WKSTN-05	T1105
7za.exe	b3a70d388488c34dd5c767692eccc9effed36b8e7c1ee03ace1bd27123a2e6d6	exfil archiver	T1560.001
PerfSvc.exe	e722dd429510c83485bb276c559015df9bd4931e7e4339eb90683cc3efd9beaa	rundll32 loader	T1218.011

Anti-forensics. p.exe survived only on **RD-01**; on file/wkstn-01/wkstn-05 it was **deleted and overwritten** (full raw-image scans after ewfmount = 0 matches; MFT entries **purged**) — a clean T1070.004, consistent with the csrss.exe timestomper.

6. Indicators of Compromise (IOCs)

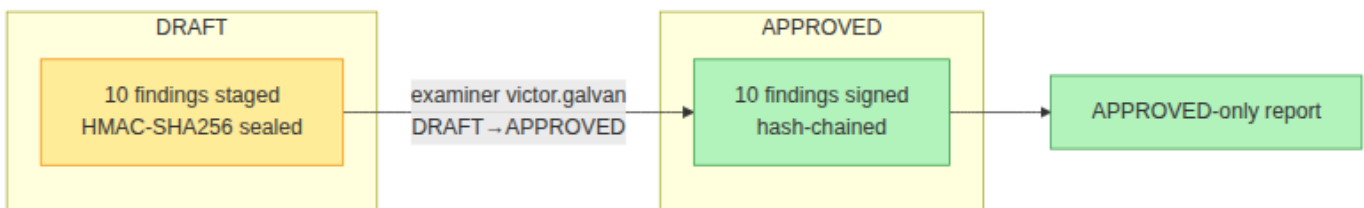
Network / behavioural



Accounts

Account	Role in intrusion
nfury	patient-zero — external RDP foothold; the IP-theft victim (Carbonadium)
tdungan	brute-forced — 692 failed NTLM from RD-01 then success
spsql	SQL service acct used for RD-01 → RD-02 RDP lateral + DC staging
rsydow-a	abused admin creds — WinRM/RDP to DC, FILE, DMZ-FTP

7. Approved findings (examiner-signed)



ID	Sev	ATT&CK	Finding
srl2018-c2-implant-001	high	T1071.004/T1055	Metasploit/Meterpreter DNS-tunnel implant p.exe (MSSE pipe) on rd-01/file/wkstn-01/05
srl2018-empire-c2-002	high	T1071.001	PowerShell Empire C2 squirreldirectory.com + localhost agents
srl2018-initial-access-003	high	T1133	nfury external RDP from 192.168.30.10/.11 → wkstn-05
srl2018-credtheft-sam-004	high	T1003.002	SAM hive stolen from DC Volume Shadow Copy
srl2018-bruteforce-005	high	T1110	692 failed NTLM brute-force of tdungan from rd-01
srl2018-persistence-006	high	T1543.003	Service persistence perfmomsv64.exe
srl2018-lateral-007	high	T1021.001/.002/.006	RD-01 ⇄ FILE lateral hub (WinRM/RDP/SMB)
srl2018-timestomp-008	med	T1070.006	Timestomping tool csrss.exe
srl2018-psexec-dmz-009	med	T1569.002	PsExec → dmz-ftp
srl2018-collection-exfil-010	high	T1560.001/T1005	Rar/7za archiving of nfury Carbonadium IP → DMZ-FTP exfil

8. Methodology, integrity & honest caveats

Approach. Agentropix-SIFT MCP toolset (68/72 tools exercised) over the SRL-2018 evidence: get_malfind/YARA recovered the in-memory implant; ewfmount+ntfs-3g mounting recovered on-disk binaries; PowerShell **4104** script-block logs yielded the attacker commands; DC **Kerberos/NTLM** (4768/4769/4776) + **RDP operational** (21/25/1149) logs built the lateral graph; prefetch confirmed execution.

Integrity. Each finding is **HMAC-SHA256 sealed** (12/12) and the approvals are **hash-chained** (DRAFT-APPROVED, examiner-signed). inference_constraint: high — the LLM orchestrated; facts come from deterministic SIFT tools.

Caveats (stated plainly).

- The eval IP 42.112.153.164 is a **synthetic test injection**, not a real lead.
- **5 of 6 memory images are smeared** (non-atomic live dc3dd/F-Response acquisition → corrupted ActiveProcessLinks) — malfind/svcscan returned 0 on those; **YARA signature scanning** (smear- immune) recovered the implant where memory-list-walking failed.
- n.ps1's full body is **not recoverable** (Empire patches AMSI / disables script-block logging post-stage); only the cradle one-liners were logged.
- Deleted p.exe copies are **unrecoverable** (content overwritten + MFT entries purged).
- evtx pulls were **capped** (8k-25k/host) — the graphs are representative, not exhaustive.
- The MCP report engine emits no single report-level seal; tamper-evidence is at the **finding** level.

Full chain of custody: docs/06-use-cases/assets/srl-2018-training-session/session-actions.log (252 steps). Samples: /home/admin2/srl-2018-malware-samples/. Tier reports (analyst/executive/ business × md/pdf/html): this directory.